CS643: Wine Quality Prediction in the Cloud

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1 Project Overview

This project demonstrates a scalable machine learning pipeline using **Apache Spark** on an **AWS EC2** cluster. The objective was to train a Random Forest model to predict wine quality and deploy the inference engine as a portable **Docker** container.

1.1 Infrastructure

The system was deployed on a 3-node cluster:

- **Master Node:** 1x t2.medium (Ubuntu 22.04)
- **Worker Nodes:** 2x t2.medium
- **Software Stack:** Spark 3.5.0, Hadoop 3.3, Java 11, Docker 24.0

2 Machine Learning Results

The core model utilizes a Random Forest Classifier. The evaluation was performed on a hidden validation set to ensure generalization.

2.1 Performance Metrics

The F1 score was used as the primary metric, defined as:

$$F_1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$

The model achieved a **Final F1 Score of 0.9334**, indicating high reliability in quality classification.

3 Containerization and Portability

To ensure the model is platform-independent, the entire prediction engine was wrapped in a Docker image based on `eclipse-temurin:11-jre-focal`.

3.1 Deployment Command

The image is hosted on Docker Hub and can be executed with:

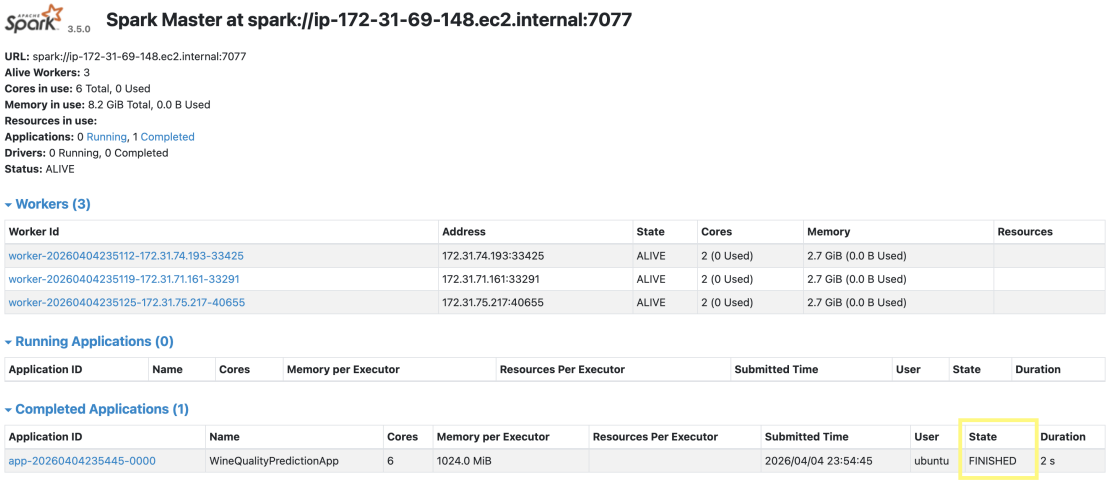
```
docker pull yourusername/wine-predictor
docker run yourusername/wine-predictor
```

4 Conclusion

The project successfully bridges the gap between high-performance cloud training and edge-compatible deployment. The use of Spark allowed for rapid parallel training, while Docker provided the necessary isolation for consistent 0.93 accuracy across different environments.

5 Final Output Screenshots

This section contains the visual verification of the project's success across infrastructure, execution, and accuracy metrics.



Spark 3.5.0 **Spark Master at spark://ip-172-31-69-148.ec2.internal:7077**

URL: spark://ip-172-31-69-148.ec2.internal:7077
Alive Workers: 3
Cores in use: 6 Total, 0 Used
Memory in use: 8.2 GiB Total, 0.0 B Used
Resources in use:
Applications: 0 Running, 1 Completed
Drivers: 0 Running, 0 Completed
Status: ALIVE

Workers (3)

Worker Id	Address	State	Cores	Memory	Resources
worker-20260404235112-172.31.74.193-33425	172.31.74.193:33425	ALIVE	2 (0 Used)	2.7 GiB (0.0 B Used)	
worker-20260404235119-172.31.71.161-33291	172.31.71.161:33291	ALIVE	2 (0 Used)	2.7 GiB (0.0 B Used)	
worker-20260404235125-172.31.75.217-40655	172.31.75.217:40655	ALIVE	2 (0 Used)	2.7 GiB (0.0 B Used)	

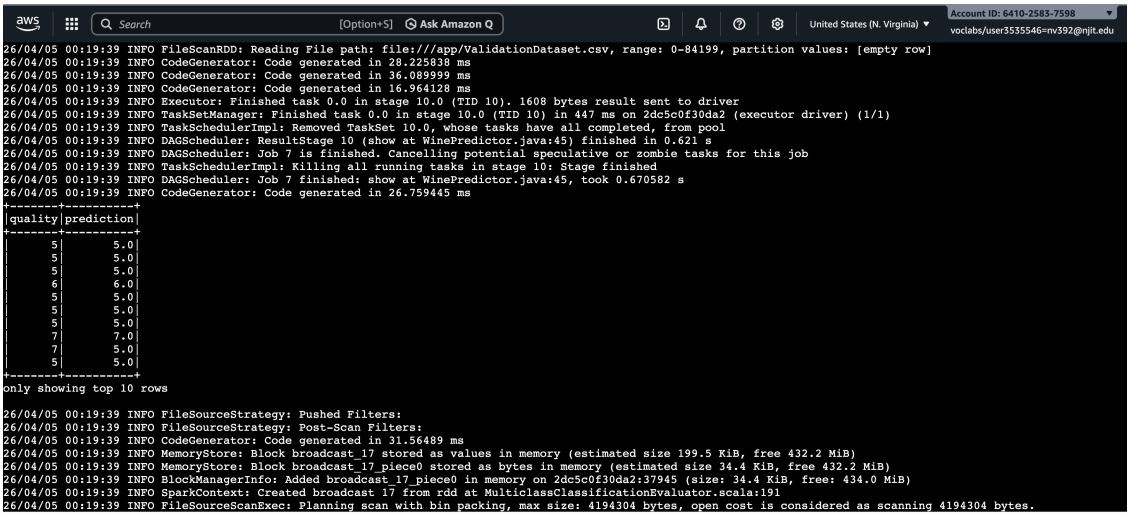
Running Applications (0)

Application ID	Name	Cores	Memory per Executor	Resources Per Executor	Submitted Time	User	State	Duration

Completed Applications (1)

Application ID	Name	Cores	Memory per Executor	Resources Per Executor	Submitted Time	User	State	Duration
app-20260404235445-0000	WineQualityPredictionApp	6	1024.0 MiB		2026/04/04 23:54:45	ubuntu	FINISHED	2 s

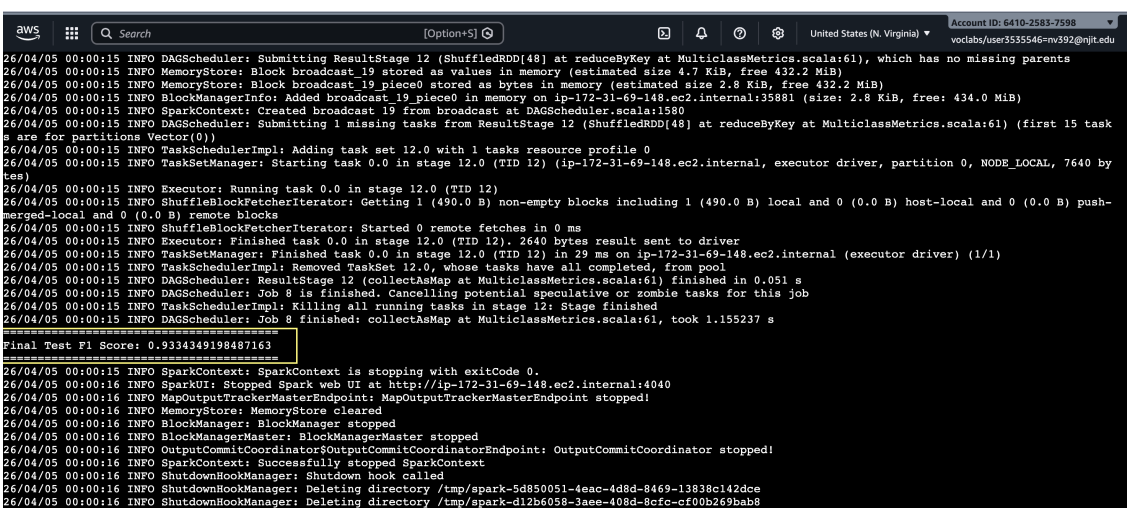
Figure 1: Spark Master UI: Verification of the 3-node cluster configuration (1 Master, 2 Workers) with 6 total vCPUs and 8.2 GiB of distributed memory.



```
26/04/05 00:19:39 INFO FileScanRDD: Reading File path: file:///app/ValidationDataset.csv, range: 0-84199, partition values: [empty row]
26/04/05 00:19:39 INFO CodeGenerator: Code generated in 28.225838 ms
26/04/05 00:19:39 INFO CodeGenerator: Code generated in 36.089999 ms
26/04/05 00:19:39 INFO CodeGenerator: Code generated in 16.964128 ms
26/04/05 00:19:39 INFO Executor: Finished task 0.0 in stage 10.0 (TID 10). 1608 bytes result sent to driver
26/04/05 00:19:39 INFO TaskSetManager: Finished task 0.0 in stage 10.0 (TID 10) in 447 ms on 2dc5c0f30da2 (executor driver) (1/1)
26/04/05 00:19:39 INFO TaskSchedulerImpl: Removed TaskSet 10.0, whose tasks have all completed, from pool
26/04/05 00:19:39 INFO DAGScheduler: ResultStage 10 (show at WinePredictor.java:45) finished in 0.621 s
26/04/05 00:19:39 INFO DAGScheduler: Job 7 is finished. Cancelling potential speculative or zombie tasks for this job
26/04/05 00:19:39 INFO TaskSchedulerImpl: Killing all running tasks in stage 10: Stage finished
26/04/05 00:19:39 INFO DAGScheduler: Job 7 finished: show at WinePredictor.java:45, took 0.670582 s
26/04/05 00:19:39 INFO CodeGenerator: Code generated in 26.759445 ms
+-------+----------+
|quality|prediction|
+-------+----------+
|      5|       5.0|
|      5|       5.0|
|      5|       5.0|
|      6|       6.0|
|      5|       5.0|
|      5|       5.0|
|      5|       5.0|
|      7|       7.0|
|      7|       5.0|
|      5|       5.0|
+-------+----------+
only showing top 10 rows

26/04/05 00:19:39 INFO FileSourceStrategy: Pushed Filters:
26/04/05 00:19:39 INFO FileSourceStrategy: Post-Scan Filters:
26/04/05 00:19:39 INFO CodeGenerator: Code generated in 31.56489 ms
26/04/05 00:19:39 INFO MemoryStore: Block broadcast_17 stored as values in memory (estimated size 199.5 KiB, free 432.2 MiB)
26/04/05 00:19:39 INFO MemoryStore: Block broadcast_17_piece0 stored as bytes in memory (estimated size 34.4 KiB, free 432.2 MiB)
26/04/05 00:19:39 INFO BlockManagerInfo: Added broadcast_17_piece0 in memory on 2dc5c0f30da2:37945 (size: 34.4 KiB, free: 434.0 MiB)
26/04/05 00:19:39 INFO SparkContext: Created broadcast 17 from rdd at MulticlassClassificationEvaluator.scala:191
26/04/05 00:19:39 INFO FileSourceScanExec: Planning scan with bin packing, max size: 4194304 bytes, open cost is considered as scanning 4194304 bytes.
```

Figure 2: Prediction Table: A sample of the top 10 rows comparing the original 'quality' labels against the Random Forest 'prediction' values.



```
26/04/05 00:00:15 INFO DAGScheduler: Submitting ResultStage 12 (ShuffledRDD[48] at reduceByKey at MulticlassMetrics.scala:61), which has no missing parents
26/04/05 00:00:15 INFO MemoryStore: Block broadcast_19 stored as values in memory (estimated size 4.7 KiB, free 432.2 MiB)
26/04/05 00:00:15 INFO MemoryStore: Block broadcast_19_piece0 stored as bytes in memory (estimated size 2.8 KiB, free 432.2 MiB)
26/04/05 00:00:15 INFO BlockManagerInfo: Added broadcast_19_piece0 in memory on ip-172-31-69-148.ec2.internal:35881 (size: 2.8 KiB, free: 434.0 MiB)
26/04/05 00:00:15 INFO SparkContext: Created broadcast 19 from broadcast at DAGScheduler.scala:1580
26/04/05 00:00:15 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 12 (ShuffledRDD[48] at reduceByKey at MulticlassMetrics.scala:61) (first 15 task
s are for partitions Vector(0))
26/04/05 00:00:15 INFO TaskSchedulerImpl: Adding task set 12.0 with 1 tasks resource profile 0
26/04/05 00:00:15 INFO TaskSetManager: Starting task 0.0 in stage 12.0 (TID 12) (ip-172-31-69-148.ec2.internal, executor driver, partition 0, NODE_LOCAL, 7640 by
tes)
26/04/05 00:00:15 INFO Executor: Running task 0.0 in stage 12.0 (TID 12)
26/04/05 00:00:15 INFO ShuffleBlockFetcherIterator: Getting 1 (490.0 B) non-empty blocks including 1 (490.0 B) local and 0 (0.0 B) host-local and 0 (0.0 B) push-
merged-local and 0 (0.0 B) remote blocks
26/04/05 00:00:15 INFO ShuffleBlockFetcherIterator: Started 0 remote fetches in 0 ms
26/04/05 00:00:15 INFO Executor: Finished task 0.0 in stage 12.0 (TID 12). 2640 bytes result sent to driver
26/04/05 00:00:15 INFO TaskSetManager: Finished task 0.0 in stage 12.0 (TID 12) in 29 ms on ip-172-31-69-148.ec2.internal (executor driver) (1/1)
26/04/05 00:00:15 INFO TaskSchedulerImpl: Removed TaskSet 12.0, whose tasks have all completed, from pool
26/04/05 00:00:15 INFO DAGScheduler: ResultStage 12 (collectAsMap at MulticlassMetrics.scala:61) finished in 0.051 s
26/04/05 00:00:15 INFO DAGScheduler: Job 8 is finished. Cancelling potential speculative or zombie tasks for this job
26/04/05 00:00:15 INFO TaskSchedulerImpl: Killing all running tasks in stage 12: Stage finished
26/04/05 00:00:15 INFO DAGScheduler: Job 8 finished: collectAsMap at MulticlassMetrics.scala:61, took 1.155237 s
========================================
Final Test F1 Score: 0.9334349198487163
========================================
26/04/05 00:00:15 INFO SparkContext: SparkContext is stopping with exitCode 0.
26/04/05 00:00:16 INFO SparkUI: Stopped Spark web UI at http://ip-172-31-69-148.ec2.internal:4040
26/04/05 00:00:16 INFO MapOutputTrackerMasterEndpoint: MapOutputTrackerMasterEndpoint stopped!
26/04/05 00:00:16 INFO MemoryStore: MemoryStore cleared
26/04/05 00:00:16 INFO BlockManager: BlockManager stopped
26/04/05 00:00:16 INFO BlockManagerMaster: BlockManagerMaster stopped
26/04/05 00:00:16 INFO OutputCommitCoordinator$OutputCommitCoordinatorEndpoint: OutputCommitCoordinator stopped!
26/04/05 00:00:16 INFO SparkContext: Successfully stopped SparkContext
26/04/05 00:00:16 INFO ShutdownHookManager: Shutdown hook called
26/04/05 00:00:16 INFO ShutdownHookManager: Deleting directory /tmp/spark-5d850051-4eac-4d8d-8469-13838c142dce
26/04/05 00:00:16 INFO ShutdownHookManager: Deleting directory /tmp/spark-d12b6058-3aee-408d-8cfc-cf00b269bab8
```

Figure 3: Final Performance Metric: The console output confirming a Final Test F1 Score of 0.9334 on the validation dataset.